

Example 1: Expand $f(z) = \frac{1}{(z-1)(z-2)}$ for $1 < |z| < 2$ by LS.

$$\rightarrow 1 < |z| < 2$$

$$\frac{1}{|z|} < 1, \quad \frac{1}{|z|} > \frac{1}{2}$$

$$\text{or, } \frac{|z|}{2} < 1$$

For convergence, expressing $f(z)$ in terms of $\frac{1}{|z|}$ or $\frac{|z|}{2}$ is needed.

$$\begin{aligned} f(z) &= \frac{1}{(z-1)(z-2)} = -\frac{1}{(z-1)} + \frac{1}{(z-2)} \\ &= -\frac{1}{z(1-\frac{1}{z})} + \frac{1}{-2(1-\frac{z}{2})} \end{aligned}$$

$$= -\frac{1}{z} \left(1 - \frac{1}{z}\right)^{-1} - \frac{1}{2} \left(1 - \frac{z}{2}\right)^{-1}$$

$$\begin{aligned} &= -\frac{1}{z} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right) - \frac{1}{2} \left(1 + \frac{1}{z} \right. \\ &\quad \left. + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right) \end{aligned}$$

Exercises : Expand $f(z) = \frac{z}{(z-1)(2-z)}$ in a LS valid for (a) $|z| < 1$, (b) $|z| > 2$,
(c) $1 < |z| < 2$, (d) $|z-1| > 1$, (e) $0 < |z-2| < 1$

$$\begin{aligned} \rightarrow \frac{z}{(z-1)(2-z)} &= \frac{1}{z-1} + \frac{2}{2-z} \\ &= \frac{1}{-(1-z)} + \frac{1}{1-\frac{z}{2}} \quad (\text{for a}) \\ &= -1(1-z)^{-1} + 1\left(1-\frac{z}{2}\right)^{-1} \\ &= -1(1+z+z^2+z^3+\dots) + 1\left(1+\frac{z}{2}+\frac{z^2}{4}+\dots\right) \end{aligned}$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

for b, $|z| > 2$, $\frac{1}{|z|} < \frac{1}{2}$

express in $\frac{1}{|z|}$

$$\frac{1}{z(1-\frac{1}{z})} + \frac{-2}{z} \cdot \frac{1}{(1-\frac{2}{z})}$$

for c , $1 < |z| < 2$

$$\frac{1}{|z|} < 1 \quad , \quad \frac{|z|}{2} < 1$$

partial fractions $\rightarrow \frac{1}{z-1} + \frac{2}{2-z}$

$$= \frac{1}{z(1-\frac{1}{z})} + \frac{1}{(1-\frac{z}{2})}$$

$$= \frac{1}{z} \left(1 - \frac{1}{z}\right)^{-1} + 1 \left(1 - \frac{z}{2}\right)^{-1}$$

for d , $|z-1| > 1$

$$\text{let, } w = z-1$$

$$z = w+1$$

$$2-z = 1-w$$

$$f(z) = \frac{w+1}{w(1-w)} = \frac{1}{w} + \frac{2}{1-w}$$

$$|z-1| > 1 \rightarrow |w| > 1 \quad \frac{1}{|w|} < 1$$

$$\frac{2}{1-w} = \frac{2}{-w} \cdot \frac{1}{\left(1 - \frac{1}{w}\right)}$$

$$f(z) = \frac{1}{w} - \frac{2}{w} \left(1 - \frac{1}{w}\right)^{-1}$$

$$= \frac{1}{w} - \frac{2}{w} \left(1 + \frac{1}{w} + \frac{1}{w^2} + \dots\right)$$

$$= \frac{-1}{z-1} - \frac{2}{(z-1)^2} - \frac{2}{(z-1)^3} - \dots$$

for e , $0 < |z-2| < 1$

$$\omega = z-2$$

$$z = \omega + 2$$

$$z-1 = \omega+1$$

$$2-z = -\omega$$

$$f(z) = -\frac{\omega+2}{\omega(\omega+1)}$$

$$= -\frac{2}{\omega} + \frac{1}{\omega+1}$$

$0 < |\omega| < 1$, in terms of ω .

$$\frac{1}{\omega+1} = \sum_{n=0}^{\infty} (-1)^n \omega^n$$

$$f(z) = -\frac{2}{\omega} + \sum_{n=0}^{\infty} (-1)^n \omega^n$$

$$= \frac{-2}{z-2} + \sum_{n=0}^{\infty} (-1)^n (z-2)^n$$

$$0 < |z-2| < 1$$

$$\boxed{11} \quad f(z) = \frac{z}{(z-1)(z+1)^2} \quad \text{Find Residue at poles}$$

for $z=1$, infinite, order 1

$z=-1$, " , order 2

Residue at $z=1$,

$$\lim_{z \rightarrow 1} (z-1) \left\{ \frac{z}{(z-1)(z+1)^2} \right\}$$

$$= \frac{1}{2^2} = \frac{1}{4}$$

at $z=-1$

$$\lim_{z \rightarrow -1} \frac{1}{1!} \cdot \frac{d}{dz} \left\{ \cancel{(z+1)} \cdot \frac{z}{(z-1)\cancel{(z+1)^2}} \right\}$$

$$= \lim_{z \rightarrow -1} 1 \cdot \frac{d}{dz} \left(\frac{z}{z-1} \right)$$

$$= \lim_{z \rightarrow -1} \frac{(1)(z-1) - (1)(z)}{(z-1)^2} = \frac{-1}{(-2)^2} = -\frac{1}{4}$$

Example 1: Determine the poles of $\frac{4-3z}{z(z-1)(z-2)}$, over $C: |z| = \frac{3}{2}$ and residue at each pole.

poles $\rightarrow z=0, z=1, z=2$

$C: |z| = \frac{3}{2}$, centre at origin
 $= 1.5$

encloses $z=0, z=1$, $z=2$ lies
outside
domain

Residue at $z=0$,

$$\lim_{z \rightarrow 0} (z-0)f(z)$$

$$= \lim_{z \rightarrow 0} \cancel{z} \cdot \frac{4-3z}{\cancel{z}(z-1)(z-2)} = 2$$

at $z=1$, -1 .

Example 2: Determine the poles of $\frac{4-3z}{z(z-1)(z-2)}$, over $C: |z| = \frac{3}{2}$ and residue at each pole. Hence evaluate $\int_C \frac{4-3z}{z(z-1)(z-2)} dz$, over $C: |z| = \frac{3}{2}$.

$$\int_C \frac{4-3z}{z(z-1)(z-2)} dz$$

$$= 2\pi i (\text{sum of residues at all poles within } C)$$

$$= 2\pi i (2-1) = 2\pi i$$

Example 3:

Find the residues of (a) $f(z) = \frac{z^2 - 2z}{(z+1)^2(z^2+4)}$ and (b) $f(z) = e^z \csc^2 z$ at all its poles in the finite plane.

$$(a) \quad z = -1, \quad z = \pm 2i$$

↓

$$\lim_{z \rightarrow -1} \frac{1}{1!} \cdot \frac{d}{dz} \left\{ (z+1)^2 \cdot \frac{z^2 - 2z}{(z+1)^2(z^2+4)} \right\} = -\frac{14}{25}$$

$$z = 2i,$$

$$\lim_{z \rightarrow 2i} \left\{ (z-2i) \frac{z^2 - 2z}{(z+1)^2(z+2i)(z-2i)} \right\} = \frac{7+i}{25}$$

$$z = -2i, \quad \text{Res} = \frac{7-i}{25}$$

$$(b) \quad e^z \csc^2 z = \frac{e^z}{\sin^2 z} \quad z = 0, \pm\pi, \pm 2\pi, \dots$$

$$z = n\pi$$

$$\lim_{z \rightarrow n\pi} \frac{1}{1!} \frac{d}{dz} \left\{ (z-n\pi)^2 \frac{e^z}{\sin^2 z} \right\}$$

made simpler by substituting

$$z - m\pi = u$$

$$z = u + m\pi \quad \left. \vphantom{z = u + m\pi} \right\} \text{ after differentiation}$$

$$z - m\pi \rightarrow 0, u \rightarrow 0$$

$$\lim_{u \rightarrow 0} e^{u+m\pi} \left\{ \frac{u^2 \sin u + 2u \sin u - 2u^2 \cos u}{\sin^3 u} \right\}$$

$$= e^{m\pi} \left\{ \lim_{u \rightarrow 0} \quad \right\}$$

L'Hospital's rule applied to find

$$\text{Res} = e^{m\pi}$$

✓ **Example 4:** Evaluate $\frac{1}{2\pi i} \oint_C \frac{e^{zt}}{z^2(z^2 + 2z + 2)} dz$ around the circle C with equation $|z| = 3$.

$$f(z) = \frac{e^{zt}}{z^2(z^2 + 2z + 2)} \Rightarrow \begin{matrix} z=0 \\ z^2 + 2z + 2 \end{matrix}$$

$$ax^2 + bx + c$$

$$\hookrightarrow \frac{-2 \pm \sqrt{4 - 4 \times 2}}{2 \times 1}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$z = -1 \pm i$$

Residue calculated at $z=0$ (order 2)
and $-1+i, -1-i$ (order 1)

$$\oint_C f(z) dz = 2\pi i (\sum \text{Res})$$

$$\frac{1}{2\pi i} \times 2\pi i (\sum \text{Res}) = \frac{t-1}{2} + \frac{1}{2} e^{-t} \cos t$$

✓ **Example 5:** Evaluate $\int_0^{\infty} \frac{dx}{x^6 + 1}$ $\rightarrow$ $\oint_C \frac{dz}{z^6 + 1}$

$$z^6 + 1 = 0$$

$$z^6 = -1$$

$$z^6 = i^2$$

$$z^3 = i, -i$$

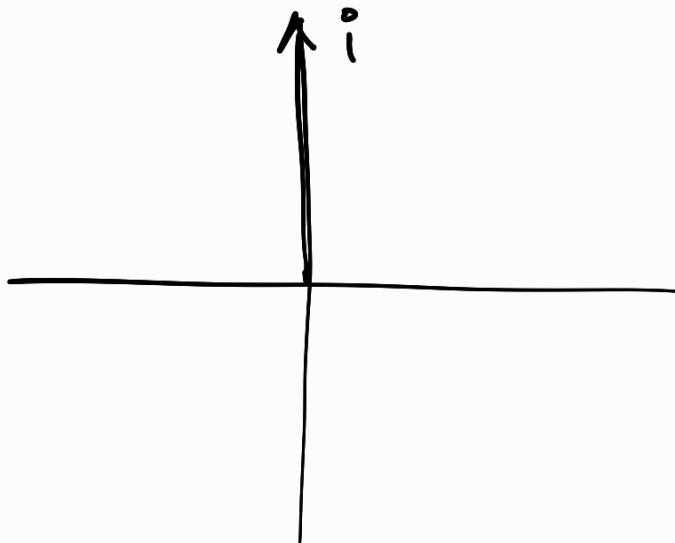
$$z = \sqrt[3]{i}, \sqrt[3]{-i}$$

Complex $\rightarrow e^{i\theta}$
num

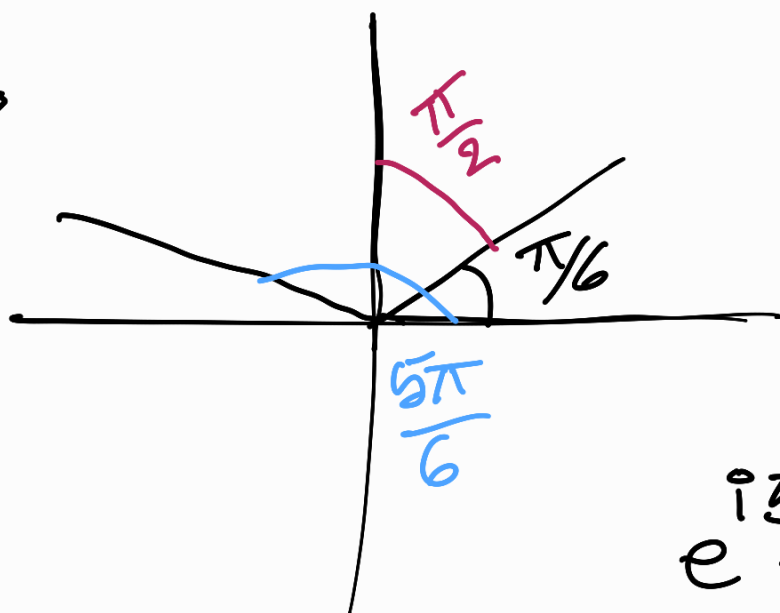
i angle $\rightarrow 90^\circ$

Cube root of i angle $\rightarrow 30^\circ$
 90°

$1, \omega, \omega^2$

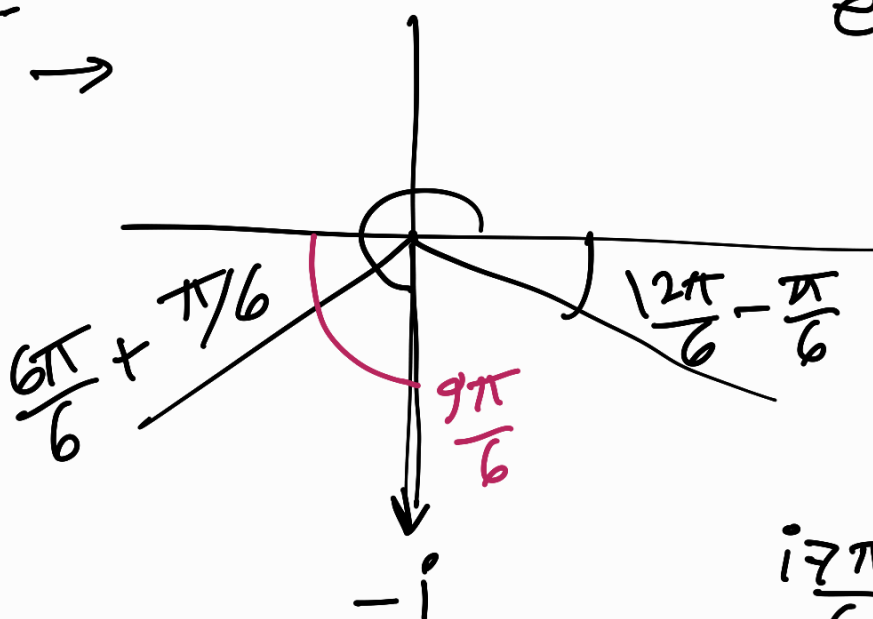


$$\sqrt[3]{i} \rightarrow$$



$$e^{i\frac{\pi}{6}}, e^{i\frac{3\pi}{6}}, e^{i\frac{5\pi}{6}}$$

$$\sqrt[3]{-i} \rightarrow$$



$$-1, -\omega, -\omega^2$$

$$e^{i\frac{7\pi}{6}}, e^{i\frac{9\pi}{6}}, e^{i\frac{11\pi}{6}}$$

$$\text{poles} \rightarrow \alpha = e^{i\frac{\pi}{6}}, e^{i\frac{3\pi}{6}}, e^{i\frac{5\pi}{6}}, e^{i\frac{7\pi}{6}}, e^{i\frac{9\pi}{6}}, e^{i\frac{11\pi}{6}}$$

Residues calc for all using L'Hospital

$$\oint_C f(z) dz = 2\pi i (\sum \text{Residue})$$

$$= \frac{2\pi}{3}$$

$$\oint_C \frac{dz}{z^6+1} \rightarrow C \text{ closed contour,}$$

consisting of the
line from $-R$ to R
and semicircle Γ
traversed in
anticlockwise

$$\int_{-R}^R \frac{dx}{x^6+1} + \int_{\Gamma} \frac{dz}{z^6+1} = \frac{2\pi}{3}$$

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{dx}{x^6+1} = \int_{-\infty}^{\infty} \frac{dx}{x^6+1} = \frac{2\pi}{3}$$

$$\int_{-\infty}^{\infty} \frac{dx}{x^6+1} = 2 \int_0^{\infty} \frac{dx}{x^6+1} = \frac{2\pi}{3} \quad [\text{even func}]$$

$$\therefore \int_0^{\infty} \frac{dx}{x^6+1} = \frac{\pi}{3}$$

Example 6: Show that $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)^2(x^2+2x+2)} = \frac{7\pi}{50}$.

Similar to 5.

Example 7: Evaluate $\int_0^{2\pi} \frac{d\theta}{3 - 2\cos\theta + \sin\theta}$.

Solution

Let $z = e^{i\theta}$. Then $\sin\theta = (e^{i\theta} - e^{-i\theta})/2i = (z - z^{-1})/2i$, $\cos\theta = (e^{i\theta} + e^{-i\theta})/2 = (z + z^{-1})/2$, $dz = iz d\theta$ so that

$$\int_0^{2\pi} \frac{d\theta}{3 - 2\cos\theta + \sin\theta} = \oint_C \frac{dz/iz}{3 - 2(z + z^{-1})/2 + (z - z^{-1})/2i} = \oint_C \frac{2 dz}{(1 - 2i)z^2 + 6iz - 1 - 2i}$$

where C is the circle of unit radius with center at the origin (Fig. 7-6).

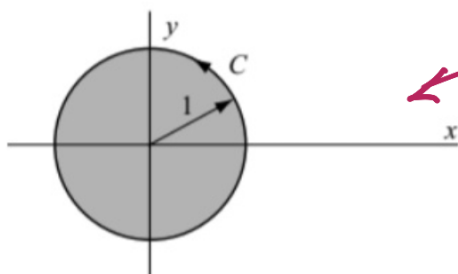


Fig. 7-6

The poles of $2/\{(1 - 2i)z^2 + 6iz - 1 - 2i\}$ are the simple poles

$$z = \frac{-6i \pm \sqrt{(6i)^2 - 4(1 - 2i)(-1 - 2i)}}{2(1 - 2i)} \\ = \frac{-6i \pm 4i}{2(1 - 2i)} = \underline{2 - i}, \underline{(2 - i)/5}$$

Only $(2 - i)/5$ lies inside C .

Residue at

$$(2 - i)/5 = \lim_{z \rightarrow (2 - i)/5} \{z - (2 - i)/5\} \left\{ \frac{2}{(1 - 2i)z^2 + 6iz - 1 - 2i} \right\} \\ = \lim_{z \rightarrow (2 - i)/5} \frac{2}{2(1 - 2i)z + 6i} = \frac{1}{2i}$$

by L'Hospital's rule.

Then

$$\oint_C \frac{2 dz}{(1 - 2i)z^2 + 6iz - 1 - 2i} = 2\pi i \left(\frac{1}{2i} \right) = \pi,$$

the required value.

De Moivre's Theorem

2π
↓
0

$$\sqrt{2^2 + 1^2} \\ \sqrt{5} > 1$$

$$\sqrt{\frac{4}{25} + \frac{1}{25}}$$

$$\sqrt{\frac{1}{5}}$$

$$\frac{1}{\sqrt{5}} < 1$$

✓ **Example 8:** Show that $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta = \frac{\pi}{12}$.

Solution

Let $z = e^{i\theta}$. Then $\cos \theta = (z + z^{-1})/2$, $\cos 3\theta = (e^{3i\theta} + e^{-3i\theta})/2 = (z^3 + z^{-3})/2$, $dz = iz d\theta$ so that

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta = \oint_C \frac{(z^3 + z^{-3})/2}{5 - 4(z + z^{-1})/2} \frac{dz}{iz} = -\frac{1}{2i} \oint_C \frac{z^6 + 1}{z^3(2z - 1)(z - 2)} dz$$

where C is the contour of Fig. 7-6.

The integrand has a pole of order 3 at $z = 0$ and a simple pole $z = \frac{1}{2}$ inside C .

Residue at $z = 0$ is

$$\lim_{z \rightarrow 0} \frac{1}{2!} \frac{d^2}{dz^2} \left\{ z^3 \cdot \frac{z^6 + 1}{z^3(2z - 1)(z - 2)} \right\} = \frac{21}{8}$$

Residue at $z = \frac{1}{2}$ is

$$\lim_{z \rightarrow 1/2} \left\{ \left(z - \frac{1}{2} \right) \cdot \frac{z^6 + 1}{z^3(2z - 1)(z - 2)} \right\} = -\frac{65}{24}$$

Then

$$-\frac{1}{2i} \oint_C \frac{z^6 + 1}{z^3(2z - 1)(z - 2)} dz = -\frac{1}{2i} (2\pi i) \left\{ \frac{21}{8} - \frac{65}{24} \right\} = \frac{\pi}{12} \text{ as required.}$$

* Always check if pole inside C or not